Beautiful Array (medium)

Problem Statement

Given a positive integer `N`, construct a beautiful array of size `N` containing the number `[1, N]`.

An array is beneficial if it follows the conditions below.

- If for any three indices `i`, `j`, `k` (with `i < j < k`), `A[j] * 2` is not equal to `A[i] + A[k].`

Examples

Example 1

- **Input**: `4`
- **Expected Output**: `[2, 1, 4, 3]`
- **Justification**: In this array, no combination of `i`, `j`, `k` (where `i < j < k`) exists such that `2 * A[j] = A[i] + A[k]`.

Example 2

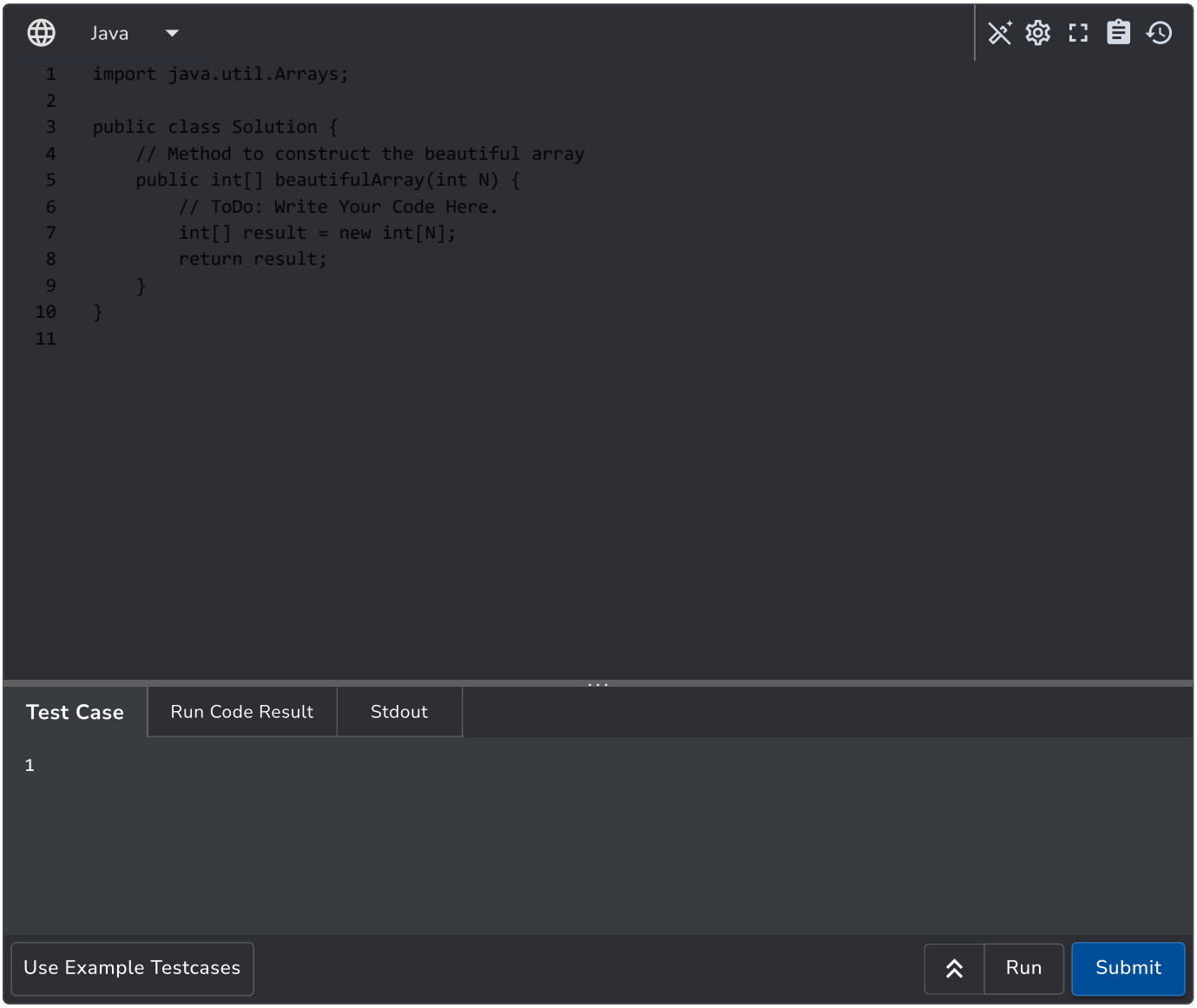
- **Input**: `3`
- **Expected Output**: `[1, 3, 2]`
- **Justification**: Similar to example 1, this array also satisfies the condition for a smaller size.

Example 3

- **Input**: `8`
- **Expected Output**: `[1, 5, 3, 7, 2, 6, 4, 8]` (or any other permutation that satisfies the condition)
- **Justification**: In this array, for every `i, j, k` (where `i < j < k`), `2 * A[j]` is never equal to `A[i] + A[k]`. This output ensures that all integers from 1 to 8 are used, and the condition is met for all possible triplets.

Try it yourself

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Solution: Beautiful Array

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